

Determinants: Formula Sheet & Reference

Class 12 Mathematics – Chapter 4 Complete Reference & Formulas

1 Evaluation of Determinants

- **1. Order 1 Determinant:**

$$A = [a] \implies |A| = a$$

- **2. Order 2 Determinant (2×2):**

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$

- **3. Order 3 Determinant (3×3): Expansion along Row 1 (R_1):**

$$\begin{aligned} |A| &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22}) \end{aligned}$$

- **4. Scalar Scaling Rule for Determinants:** If A and B are square matrices of order n such that $A = kB$, then:

$$|A| = |kB| = k^n |B|$$

2 Coordinate Geometry Application: Area of Triangle

- **Area of Triangle with Vertices** $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Properties:

- Since area is always positive, take the absolute value $|\Delta|$.
 - If area is given, use $\pm\Delta$ to solve for unknown coordinates.
 - Three points are **collinear** if and only if $\Delta = 0$.
- **Equation of Line Passing Through** (x_1, y_1) **and** (x_2, y_2) :

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

3 Minors, Cofactors & Expansion Properties

- **Minor** (M_{ij}): Determinant of submatrix obtained by deleting i -th row and j -th column.
- **Cofactor** (A_{ij}):

$$A_{ij} = (-1)^{i+j} M_{ij}$$

- **Sum of Products with Own Row/Column Cofactors:**

$$|A| = \sum_{j=1}^3 a_{ij} A_{ij} = a_{i1} A_{i1} + a_{i2} A_{i2} + a_{i3} A_{i3}$$

- **Sum of Products with Other Row/Column Cofactors (Orthogonality):**

$$\sum_{j=1}^3 a_{ij} A_{kj} = a_{i1} A_{k1} + a_{i2} A_{k2} + a_{i3} A_{k3} = 0 \quad (\text{for } i \neq k)$$

4 Adjoint & Inverse of a Square Matrix

- **Adjoint of Matrix A ($\text{adj } A$):**

$$\text{adj } A = [A_{ij}]^T = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

Shortcut for 2×2 Matrix: Interchange diagonal elements, change signs of off-diagonal elements:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies \text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- **Fundamental Adjoint Identity (Theorem 1):**

$$A(\text{adj } A) = (\text{adj } A)A = |A|I_n$$

- **Determinant of Products & Adjoint Identities:**

$$\begin{aligned} |AB| &= |A||B| \\ |\text{adj } A| &= |A|^{n-1} \quad (\text{for } n \times n \text{ matrix}) \\ |A^{-1}| &= \frac{1}{|A|} = (\det A)^{-1} \end{aligned}$$

- **Singular vs Non-Singular Matrices:**

- **Singular:** $|A| = 0$ (Non-invertible).
- **Non-Singular:** $|A| \neq 0$ (Invertible).

- **Matrix Inverse Formula (Theorem 4):**

$$A^{-1} = \frac{1}{|A|}(\text{adj } A) \quad (\text{exists} \iff |A| \neq 0)$$

5 System of Linear Equations (Matrix Method)

- **Matrix Equation Representation ($AX = B$):**

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

- **Criteria for Consistency & Solutions:**

- **Case 1 ($|A| \neq 0$):** System is **consistent** with a unique solution:

$$X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$$

- **Case 2 ($|A| = 0$ and $(\text{adj } A)B \neq O$):** System is **inconsistent** (No solution).
- **Case 3 ($|A| = 0$ and $(\text{adj } A)B = O$):** System may be **consistent** (infinitely many solutions) or **inconsistent** (no solution).